

Supplementary Materials: Spatial Context Causal Adjustment for Geographic Observational Studies

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1 S1. Proof of Proposition 1

Recall the data-generating process (DGP) in the main text:

$$Y_i = \tau T_i + X_i^\top \gamma + C_i^\top \phi + \sigma_u U_i + \nu_i, \quad U = (I - \rho W)^{-1} \epsilon, \quad \epsilon \sim \mathcal{N}(0, I_n), \quad \nu \sim \mathcal{N}(0, \sigma_\nu^2 I_n), \quad (1)$$

with $|\rho| < 1$, W row-standardised and symmetric in expectation (its eigenvalues lie in $[-1, 1]$), and $\nu \perp U$.

Let $Z = [\mathbf{1}, T, X, C]$ and $M = I - Z(Z^\top Z)^{-1}Z^\top$ be the residual projector for the SCCA-adjusted model. The fitted SCCA residual is

$$r = MY = M(\sigma_u U + \nu), \quad (2)$$

because $MZ\beta = 0$ for any coefficient vector β . The row-standardised Moran's I computed on r is

$$I(r) = \frac{n}{s_0} \cdot \frac{r^\top W r}{r^\top r}, \quad s_0 = \mathbf{1}^\top W \mathbf{1} = n \quad (\text{row standardisation}), \quad (3)$$

so $n/s_0 = 1$ and $I(r) = r^\top W r / r^\top r$.

Numerator. Substituting (??) into the numerator of (??),

$$r^\top W r = \sigma_u^2 U^\top M W M U + 2\sigma_u U^\top M W M \nu + \nu^\top M W M \nu. \quad (4)$$

Because $\nu \perp U$ and $E[\nu] = 0$, the cross term vanishes in expectation, and

$$E[\nu^\top M W M \nu] = \sigma_\nu^2 \cdot \text{tr}(M W M). \quad (5)$$

For U , write $\Sigma_U = (I - \rho W)^{-1}(I - \rho W^\top)^{-1}$. Then $E[U U^\top] = \Sigma_U$ and

$$E[U^\top M W M U] = \text{tr}(M W M \Sigma_U). \quad (6)$$

If W is symmetric or close to symmetric (which holds for many row-standardised k-nearest-neighbor graphs after explicit symmetrisation), Σ_U has the eigenvalue expansion

$$\Sigma_U = \sum_k \frac{1}{(1 - \rho \lambda_k)^2} v_k v_k^\top, \quad (7)$$

where $\{\lambda_k, v_k\}$ are the eigenpairs of W . Defining $P = M W M$, by the Loewner ordering on the trace and using $\lambda_k \in [-1, 1]$, the diagonal terms with $\lambda_k > 0$ dominate when $\rho > 0$, giving

$$\text{tr}(M W M \Sigma_U) \geq \frac{\rho}{(1 - \rho)^2} \cdot \text{tr}(M W M) + O(1). \quad (8)$$

Substituting (??) back gives

$$E[r^\top W r] \geq \sigma_u^2 \cdot \frac{\rho}{(1 - \rho)^2} \cdot \text{tr}(M W M) + \sigma_\nu^2 \cdot \text{tr}(M W M). \quad (9)$$

Denominator. By the same logic,

$$E[r^\top r] = \sigma_u^2 \cdot \text{tr}(M \Sigma_U M) + \sigma_v^2 \cdot \text{tr}(M) \leq (\sigma_u^2 + \sigma_v^2) \cdot \text{tr}(M) \cdot \frac{1}{(1 - |\rho|)^2}. \quad (10)$$

Lower bound on $I(r)$. Combining numerator and denominator, with a Taylor expansion in the Moran ratio that contributes only $o(n^{-1})$ when both $\text{tr}(MWM)$ and $\text{tr}(M)$ grow linearly with n (the usual rate for projection matrices), we obtain

$$E[I(r)] \geq \frac{\sigma_u^2}{\sigma_u^2 + \sigma_v^2} \cdot \frac{\rho}{(1 - \rho)^2} \cdot \kappa(W, M) - o(n^{-1}), \quad (11)$$

where

$$\kappa(W, M) := \frac{\text{tr}(MWM)}{\text{tr}(M)} \cdot (1 - |\rho|)^2 \in (0, 1]. \quad (12)$$

This is exactly equation (5) in the main text. $\square$

Remarks. (1) The bound is monotone in $|\rho|$ for $\rho \in [0, 1)$ and monotone in $\sigma_u^2/(\sigma_u^2 + \sigma_v^2)$. (2) $\kappa(W, M) \rightarrow 1$ as the projection M becomes orthogonal to the eigenvectors corresponding to negative eigenvalues of W , and falls toward 0 when M aligns with those eigenvectors. (3) The bound does not identify ρ or σ_u ; it only quantifies how large the residual Moran statistic must be if a SAR confounder of given strength is responsible for it.

2 S2. Threshold-calibration DGP

The calibration script in `scripts/threshold_calibration.py` realises this DGP with the following choices:

- Coordinates: $\sqrt{n} \times \sqrt{n}$ jittered grid in $[0, 1]^2$.
- Neighborhood graph W : 8-nearest-neighbour, row-standardised.
- Latent confounder: $U = (I - \rho W)^{-1} \epsilon$ with $\epsilon \sim \mathcal{N}(0, I)$.
- Observed covariate: $x \sim \mathcal{N}(0, 1)$; observed terrain proxy: $\text{elev} \sim \mathcal{N}(0, 1)$.
- Treatment: logistic with linear predictor $0.5x + 0.5 \cdot \text{elev} + \sigma_u U$.
- Outcome: $y = \tau T + x + 0.7 \text{elev} + \sigma_u U + 0.5 \eta$, $\eta \sim \mathcal{N}(0, 1)$ with $\tau = 0.5$.
- SCCA-adjusted estimator: linear regression of y on $(1, T, x, \text{elev})$.
- Spatial sensitivity: SLX-style regression on $(1, T, WT, x, Wx, \text{elev})$.
- Sweep: $\rho \in \{0.2, 0.5, 0.8\}$, $n \in \{500, 2000, 5000\}$, $\sigma_u \in \{0, 0.5, 1, 2\}$, 50 replicates per cell.

3 S3. Output contracts

Every SCCA run writes the following files when feasible:

- `manifest.json` – inputs, exposure/outcome fields, variant set, seed, threshold rule version.
- `effect_estimates.csv` – one row per estimator.
- `erf_curve.csv` – exposure grid and fitted response.
- `balance_summary.csv` – pre/post SMDs per covariate.
- `overlap_summary.json` – common-support attrition.

- `spatial_robustness.csv` – residual Moran, neighbour-exposure, SLX summary, graph sensitivity.
- `bootstrap_robustness.csv` – block bootstrap replicates.
- `placebo_tests.csv` – placebo exposure thresholds.
- `credibility_report.json` – machine-readable evidence grade.

GIS-aware runs additionally write `*.gpkg`, `*.geojson`, `*.shp` and `*.qml` style files keyed to the same `AnalysisRequest` call.